1.1.1 Pipeline Map

NOTE: This pipeline will be run on the full sample, as well as both sex-stratified samples

Required Inputs

1. **Phenotype/Design Matrix** — Phenotype and control covariates
2. **Omics** — Omics dataset

Input Structure: Phenotype/Design Matrix

Required Columns

Column	Description	Valid Values
SAMPLE_ID	Sample identifier	Unique throughout the pheno/design matrix
FU	Follow-up timepoint	0 (baseline), 1 (first follow-up), 2 (second follow-up, if applicable)
SUBJECT_ID	Subject-specific identifier	SUBJECT_ID/FU pairs must be unique
FEMALE	Female status	0 or 1
CONTROL_STATUS	Control status	0 or 1

Additional Covariates

Other covariates must meet the following requirements:

- **Specification**: Via character vector provided as additional input, specifying the columns that represent these covariates
- **Allowed Types**: Numeric, Factor, or Logical
- **Data Quality**: (Possibly) No NAs? No near zero variance (NZV)?

Per-Variable Auto-Report

For each covariate, generate:

- **Variable type**
- **Summary statistics**:
    - If numeric: summary() output
    - If factor: Number of levels, level names, sample size per level
    - If logical: Sample size per logical value

Input Structure: Omics Matrix

Format Requirements

- **Orientation**: Rows are analytes, columns are samples
- **Sample IDs**: Column names correspond to SAMPLE_ID from pheno matrix
- **Analyte Names**: Can be provided as either:
    - Row names, OR
    - In an analyte column

(Either format is acceptable)

Quality Checks

- Report overlap/missingness between omics and pheno matrices
- Ensure there are no replicates (Same Subject ID at same follow-up – average them)

Validation and Reporting

Pheno/Design Matrix Validation

Variable	Validation	Reporting
SAMPLE_ID	All values unique	N
FU	Unique values are only 0, 1, and (possibly) 2	N of each level
SUBJECT_ID	All SUBJECT_ID/FU pairs are unique	N unique subjects
FEMALE	Unique values are only 0 and 1	N of each level
CONTROL_STATUS	Unique values are only 0 and 1	N of each level

Other Covariates	See "Additional Covariates" requirements above	See "Per-Variable Auto-Report" above

Omics Matrix Validation

Validation Checks:

- Column names overlap with SAMPLE_ID from pheno matrix (report total common overlap)
- All columns are numeric
- No columns with all NAs or NZV (possibly)

Reporting:

- Number of analytes
- Number of samples (before and after merging with pheno data)
- Per-analyte summaries (to accompany results):
    - Number of NAs at each follow-up
    - Mean, median, range, SD

Computation

Data Processing

1. **Preprocessing** (if applicable):
    a. **Scaling**: Yes
    b. **Skewness handling**: Maybe
    c. **Batch correction**: No (likely)
2. **Data Preparation**:
    a. For non-limma models: Merge pheno and omics data objects into a single matrix
    b. Transpose omics matrix as needed to align samples
3. **Model Selection**:
    a. Determine whether to use lm, lme4, or limma
    b. **DNAm data**: Data type will be a flag, with no default

   c. **Model type**:
      i. Use mixed effects if FU contains value 2
      ii. Use linear model (lm) otherwise
4. **Hypothesis Testing**:
   a. Apply appropriate hypothesis test using selected package
   b. Aggregate results across all analytes
5. **Multiple Testing Correction**:
   a. Calculate Bonferroni-adjusted p-values
   b. Calculate Benjamini-Hochberg (BH) adjusted p-values

Outputs

1. Results File (CSV)

Primary analysis results with the following columns:

Column	Description
ANALYTE_NAME	Analyte identifier
EFFECT_SIZE	Estimated effect size
SE	Standard error of estimate
P_VALUE	Unadjusted p-value
BH_P_VALUE	Benjamini-Hochberg adjusted p-value

2. Omics Summary (CSV)

Column	Description
ANALYTE_NAME	Analyte identifier
N_NONMISSING	Number of non-missing observations
MEAN	Mean analyte value
MEDIAN	Median analyte value
SD	Standard deviation
MIN	Minimum value
MAX	Maximum value

3. Phenotype Summary (JSON)

Summary statistics for the phenotype/design matrix:

```
{
  "N_SAMPLES": <total_samples>,
  "N_SAMPLES_FU0": <samples_at_baseline>,
  "N_SAMPLES_FU1": <samples_at_fu1>,
  "N_SAMPLES_FU2": <samples_at_fu2>, // Depending on the dataset
  "N_SUBJECTS": <unique_subjects>,
  "N_FEMALE": <female_count>,
  "N_MALE": <male_count>,
  "N_CONTROL": <control_count>,
  "N_TREATMENT": <treatment_count>
}
```

4. 'Other' Covariates Summary (CSV)

Summary of additional covariates. Each covariate is a top-level key with nested summary statistics appropriate to its type.

JSON Structure:

```
{
  "<covariate_name>": {
    "type": "numeric|factor|logical",
    "n_na": <count>,
    "summary": {
      // Type-specific summary (see below)
    }
  }
}
```

Summary field format:

- If numeric: Numerical summary (mean/median/sd/range)
- If factor: Number of levels and sample size per level
- If logical: Sample size per logical value

Notes and Decisions Needed

- **NZV and NA handling**: Confirm whether strict "no NAs" and "no NZV" requirements should be enforced
- **Analyte summaries**: Confirm final format for per-analyte quality metrics
- **Omics reporting**: Determine if Report 3 is needed as a separate output or if validation-stage reporting is sufficient
- **Model selection flags**: Confirm whether DNAm detection should be automatic or flag-based